

Application of Trigonometry

Question:

A 50 m long tree breaks during a storm, and its broken part touches the ground. The larger broken part is 8 m longer than the smaller part.

A boy is standing at the top of a 100 m high tower. The distance between the tower and the point where the tip of the broken tree touches the ground is twice the distance between that point and the root of the tree.

If the angle of depression of the root of the tree from the boy's eye is 60° , find the height of the boy.

Answer

Correct Answer: 3.92 m

Diagram

